# Live Percentage Sheet

| | |
|---|---|
| $\frac{1}{2} = 50\%$ | $\frac{1}{8} = 12.50\%$ or $12\frac{1}{2}\%$ |
| $\frac{1}{3} = 33.33\%$ or $33\frac{1}{3}\%$ | $\frac{1}{9} = 11.11\%$ or $11\frac{1}{9}\%$ |
| $\frac{1}{4} = 25\%$ | $\frac{1}{11} = 9.09\%$ or $9\frac{1}{11}\%$ |
| $\frac{1}{5} = 20\%$ | $\frac{1}{12} = 8.33\%$ or $8\frac{1}{3}\%$ |
| $\frac{1}{6} = 16\frac{2}{3}\%$ | $\frac{1}{14} = 7.14\%$ or $7\frac{1}{7}\%$ |
| $\frac{1}{7} = 14.28\%$ or $14\frac{2}{7}\%$ | $\frac{1}{15} = 6.66\%$ or $6\frac{2}{3}\%$ |

NOTE :

$\frac{1}{3} = 33\frac{1}{3}\% \Rightarrow \frac{2}{3} = 66\frac{2}{3}\%$

$\frac{1}{8} = 12.5\% \Rightarrow \frac{3}{8} = 37.5\%$

## DAY 01 :

### Model 1: Basic Percentage

**Concept :** Difference between 54% of a number and 63% of the same number is 72. What is 80% of that number?

TITA

1. Sameer spends 20% of his monthly salary on house-rent, 25% on food, 10% on transportation, 15% on education of his children and 18% on household expenses. He saves the remaining amount of ` 4800. What is his monthly salary? 1) ` 40000 2) ` 28000 3) ` 32000 4) ` 36000 5) None of these

2. Ishan spent ` 35,645 on buying a bike, ` 24,355 on buying a television, and the remaining 20% of the total amount he had as cash with him. What was the total amount? ` 60,000 2) ` 72,000 3) ` 75,000 4) ` 80,000 5) None of these

3. Sujatha invests 7% i.e. ` 2170, of her monthly salary in mutual funds. Later she invests 18% of her monthly salary in recurring deposits. Also she invests 6% of her salary on NSC's. What is the total annual income invested by Sujata?
1) ` 125320 2) ` 113520 3) ` 135120 4) ` 115320 5)None of these

4. Sonali invests 15% of her monthly salary in insurance policies. She spends 55% of her monthly salary in shopping and on household expenses. She saves the remaining amount of ` 12750. What is Sonali's monthly income? 1) ` 42,500 2) ` 38,800 3) ` 40,000 4) ` 35,500 5) None of these

5. Gaurav spends 40% of the amount he received from father on hostel expenses, 20% on books and stationery and 50% of the remaining on transport. He saves ` 450 which is half the remaining amount after spending on hostel expenses, books and stationery and transport. How much money did he get from his father?
1) ` 3000 2) ` 6000 3) ` 4500 4) ` 5000 5) None of these

6. Prerna decided to donate 30% of her monthly income to an orphanage. But on the day of donation, she changed her mind and donated ` 4200, which is 70% of what she had decided earlier. What should be the actual donation as per her earlier decision?

1) ` 8000 2) ` 5000 3) ` 6,000 4) Cannot be determined 5) None of these

7. Praveen decided to donate 15% of his salary to a charity. On the day of donation he changed his mind and donated ` 1,896, which was 80% of what he had decided earlier. How much is Praveen's salary? 1) ` 18,500 2) ` 10,250 3) ` 15,800 4) Cannot be determined 5) None of these

## Model 2 : A Is What Percent of B

Concepts :

**2. : x% of y is y% of?**

(A) X (B) x/100 (C) 100x (D) y/100

3. In an examination, Ramesh scored 30% less than Suresh and Mahesh scored 20% less than Suresh. Ramesh's score is what per cent of Mahesh's score?
1) 80% 2) 70% 3) 40% 4) 87.5% 5) None of these

4. 65 is what % of 50?
1) 250 2) 240 3) 160 4) 130 5) None of these

5. In a test consisting of 300 questions, Deepika answered 40% of the first 100 questions correctly. What per cent of the remaining 200 questions does she need to answer correctly for her grade on the entire exam to be 50%?
1) 75% 2) 55% 3) 60% 4) Cannot be determined 5) None of these

## Model 3: A Is What Percent More/Less than B

1. Percentage Change $= \frac{\text{Final} - \text{Initial}}{\text{Initial}} \times 100$

2. Quantity I is how much percent of Quantity II
$\frac{\text{Quantity I}}{\text{Quantity II}} \times 100$

3. Quantity I is how much percent more than Quantity II
$\frac{\text{Quantity I} - \text{Quantity II}}{\text{Quantity II}} \times 100$

4. Quantity I is how much percent less than Quantity II
$\frac{\text{Quantity II} - \text{Quantity I}}{\text{Quantity II}} \times 100$

1. If A's salary is 25% more than B's salary, then by what per cent is B's salary less than A's salary?
1) 25% 2) 20% 3) 16.66% 4) Cannot be determined 5)None of these

2. If Arjun's salary is 20% more than that of Bheem, then how much percent is Bheem's salary less than that of Arjun?
1) 16.66% 2) 20% 3) 40% 4) 10% 5) None of these

3. The sale of Company N is 40% less than that of Company T. Then by what per cent is the sale of company T more than that of N?
1) 66% 2) 20% 3) 40% 4) 10% 5) None of these

4. If A's salary is 25% less than B's salary, then B's salary is what % more than A's salary?
1) Cannot be determined 2) 21 3) 33.33 4) 25 5) 20

**Model 4: Type : Pass Fail**

5. A student has to secure 40% marks to pass an examination. He secured 75 marks and was declared FAIL 45 marks. Find the maximum marks set for the examination.
1) 600 2) 200 3) 100 4) 300 5) None of these

6. In an examination, the cut off mark is 40% and a student is declared to be failed by 4 marks when he scored 38%. What is the maximum score a student can get in this exam?
1) 100 2) 300 3) 400 4) 200 5) None of these

7. A student has to secure 30% marks to get through. If he gets 40 marks and fails by 20 marks, find the maximum marks set for the examination.
1) 600 2) 200 3) 100 4) 300 5) None of these

# DAY 02

**Model 5: Population based**

Q 1. The value of lathe machine depreciates at the rate of 10 % per annum. If the cost of machine at present is Rs. 160,000, then what will be its worth after 2 years?
a. Rs. 122,365 b. Rs. 153,680 c. Rs. 129,600 d. Rs. 119,900

Q.2 The value of a washing machine depreciates at the rate of 10% every year. If its present value is Rs. 8748, then what was the price of the washing machine three years ago?

**TITA**

Q.3 The population of different trees in a field increased by 10 % in first year, increased by 8 % in second year and decreased by 10 % in third year. If at present the number of trees is 26730, then find the number of tress in the beginning.
a. 30000 b. 25000 c. 27000 d. 27865

Q3 : The population of a town was 48600. It increased by 25% in the first year and decreased by 8% in the second year.What will be the population of the town at the end of two years?

**TITA**

## Model 6: Election Problem

1. In an election between two candidates, the winner secured 58% of the total votes cast and wins by a majority of 2400 votes. How many votes did the loser get?
1) 6300 2) 7200 3) 3700 4) 4400 5) None of these

2. In an election between two candidates, one gets 72% of the total votes. If the total votes are 8200, by how many votes did the winner win the election?
1) 1835 2) 1722 3) 3608 4) 4428 5) None of these

3. In a college election between two students, 10% of the votes cast are invalid. The winner gets 70% of the valid votes and defeats the loser by 1800 votes. How many votes were totally cast?
1) 1800 2) 7200 3) 5000 4) 3600 5) None of these

4. In a college election fought between two candidates, one candidate got 55% of the total valid votes. 15% of the votes were invalid. If the total votes were 15,200, what is the number of valid votes the other candidate got?
1) 7106 2) 6840 3) 8360 4) 5814

## Model 7: Effective Percentage Change

Q.1 : The price of an article rose by 25% every odd year and fell by 20% every even year , what would be the percentage change after 180 years ?

Q2 : A number is increased by 20% and then decreased by 20%, the final value of the number is ?

(1) increase by 2% (2) decrease by 3%. (3) decrease by 4% (4) increase by 5% (5) None of these

**3.** The length of a rectangle increased by 40% and its breadth increased by 20%. What will be the percentage change in the area of the rectangle?

(1) 50% inc (2)10% inc (3) 44% inc. (4) 68% inc (5) None of these

**4.** When the cost of petroleum increases by 40%, a man reduces his annual consumption by 20%. Find the percentage change in his annual expenditure on petroleum.

(1) 20% (2) 16% (3) 12% ( 4) 40% (5) None of these

**5.** The price of sugar increased by 25%. What should be the percentage decrease in the consumption of sugar by a family, such that their expenditure on sugar remains the same?

(1) 20% (2) 16% (3) 12% (4) 40% (5) None of these

(1) 10 (2) 20 (3) 30 (4)40. (5) None of these

**7.**What is the % change in the area of a rectangle when its length increases by 10% and its width decreases by 10%?

(1)0% (2) 20% inc (3)20% dec(4)1% dec (5) insufficient data

# Day 03

## Model 8:

### Concept: Expenditure, Price, Consumption

1. **If price of DESI-liquor is increased by 20% then Mohan should cut down consumption by what %, to keep his budget unchanged?**

2. **Dish TV has reduced the channel prices by 20% now I'm able to subscribe to 5 more channels in the same budget of Rs.400. How many channels did I subscribe earlier?**

3. **The price of sugar increases by 20% due to this a family purchase 12kg less sugar for Rs300.find**

**A. Original consumption** **B. Original price**

**C. New consumption** **D. New price**

**4.** The price of sugar having gone down by 10%, a consumer can buy 5 kg more sugar for Rs 270. The difference between the original and reduced price per kg is:

**A. 62 paise** **B. 60 paise** **C. 75 paise** **D. 53 paise**

## Miscellaneous- New Type – TITA ( Type In The Answer)

Q1**:** In an election between two candidates, 75% of the voters cast their votes, out of which 2% votes were declared invalid. A candidate got 9261 votes which were 75% of the valid votes. The total number of voters enrolled in that election was:

**TITA**

Q2: There are two candidates P and Q in an election. During the campaign, 40% of the voters promised to vote for P, and rest for Q. However, on the day of election 15% of the voter went back on their promise to vote for P and instead voted for Q. 25% of the voters went back on their promise to vote for Q and instead voted for P. P lost by 300 votes, then what was the total no. of voters.

**TITA**

Q3: Priyanka bougth a box of chocolates. She has eaten 1/4 th of them. 33.33% of the remaining chocolates she given to her friends. She gave 20% of the remaining chocolates to some kids. If she is left with 48 chocolates, how many chocolates were there in the box?

**TITA**

Q4: A man spent 16. 66% of his monthly earning on house rent. Out of the balance, he spent 12.5% on clothes, out of remaining 25% on travelling, out of the balance 22.22% on education, from remaining 27.27% on food. If he had a balance of Rs.1225 at the end of the month, the monthly earning of the man (in rupees) is

**TITA**

Q5: In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. If he attempts all 60 questions and secures 130 marks, the number of questions he attempts correctly is :

**TITA**

Q6: In an examination, P scored 30 % marks and failed by 15 marks. Q scored 40 % marks and obtained 35 marks more than those required to pass. Find the pass percentage.

**TITA**

Q7: Priyanka bougth a box of chocolates. She has eaten 1/4 th of them. 33.33% of the remaining chocolates she given to her friends. She gave 20% of the remaining chocolates to some kids. If she is left with 48 chocolates, how many chocolates were there in the box?

**TITA**

Q8: A man spent 16. 66% of his monthly earning on house rent. Out of the balance, he spent 12.5% on clothes, out of remaining 25% on travelling, out of the balance 22.22% on education, from remaining 27.27% on food. If he had a balance of Rs.1225 at the end of the month, the monthly earning of the man (in rupees) is:

**TITA**

Q9: If a student multiplied a no. By 7/12 instead of 11/16 find the % error in the calculation.

**TITA**

Q.10: Mr. Jones gave 40% of the money he had, to his wife. He also gave 20% of the remaining amount to each of his three sons. Half of the amount now left was spent on miscellaneous items and the remaining amount of Rs. 12,000 was deposited in the bank. How much money did Mr. Jones have initially?

**TITA**